

NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for policy evaluation and feasible improvement in controlled economies. Centered continuation errors connect economic transport and full-policy loss. A deterministic own-future ReLU construction carries the feasible actions attaining its labels through the minimum circuit. The resulting policy admits a one-sided residual certificate and a smaller primitive resource budget than a separated critic-actor account. Exact witness compilation and an explicit numerical-selection allowance make that guarantee executable. Feasible-witness transport extends the construction to endogenous action constraints with an explicit repair budget. A predeclared matched study compares complete certificate frontiers with strengthened neural and spline baselines. Retained-policy verification and direct expected-cost comparisons remain distinct. The evidence separates sharper certification from work-to-target and policy-cost superiority. Controlled-diffusion, recursive-preference and game formulations retain their application-specific conditions.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Recursive utility, Computational economics.

1. INTRODUCTION

An economic decision uses a continuation value because its consequences extend beyond the current period. A learned continuation is useful only to the extent that the decisions made with it are accurate. Prediction loss, a Bellman residual at sampled states, and the value of the policy actually implemented are different objects. Their distinction is particularly important when a counterfactual changes future policies or when a recursive preference evaluates the continuation generated by those policies rather than the optimum.

This paper develops Neural Bellman Operators (NBOs) as a policy-evaluation and feasible-improvement method. An operator takes economic primitives, a current policy, and computational resources as inputs. It returns a continuation, a feasible policy, and an economic error account, or it records that the requested account was not obtained within its cap. The continuation is evaluated against its own fitted future. Current actions, policy-specific future values, numerical execution, and the cost of verification enter the same account. The construction is not a proposal to rename an arbitrary neural predictor as a Bellman operator.

There are four parts to the argument. First, only action-dependent continuation errors affect a current decision. Their oscillation, rather than their absolute level, controls loss. A transport bound identifies when an anchor continuation remains valid after the future economy changes. These results connect reuse and refresh decisions to economic tolerances. They also preserve the original formulation in controlled diffusions, recursive preferences, endogenous preferences, and games, subject to the distinct hypotheses of those applications.

Second, a signed residual certificate controls a complete returned policy. Its upper bound contains the widths of the residual enclosures, the suboptimality of the selected actions, and the terminal enclosure. It is invariant to additive date-specific shifts of the fitted continuation. This leads to a useful separation: a policy can be accurate even though a coarse verifier does not establish its accuracy. We prove that residual verification can be refined while retaining the original network, the original deployed actor, and its original allowance. The resulting bound measures improved knowledge of the same policy, not a policy improvement disguised as additional verification.

Third, certification of each policy relative to the optimum does not rank two policies. We construct their cost difference directly. Under expected-cost preferences, an own-continuation Bellman telescoping identity produces an unbiased policy-cost score with a support width governed by the signed certificate. Coupling innovations across policies and enclosing every simulated path gives simultaneous finite-sample intervals for the difference in the original continuous-innovation economy. The statistical assertion is conditional on frozen policies and the stated independent

simulation model; outward arithmetic and unresolved action-cell boundaries have their own explicit treatment.

Fourth, a deterministic neural construction connects primitive moduli to a complete policy tolerance. A minimum of Lipschitz cones, realized exactly by ReLU gates, fits each date’s own-future Bellman labels without inflating its Lipschitz bound when those labels are inexact. The state, action, and arithmetic budgets are specified before construction. This yields finite termination for a defined neural backend rather than a claim about arbitrary nonlinear training. A new full matched catalogue executes the same resolution ladder for this backend and a conventional spline, including every unsuccessful rung and three isolated timing repetitions.

The action that attains a Bellman label supplies additional information. We retain its index through the ReLU minimum circuit and deploy that action when its cone is selected. A one-sided policy sandwich then avoids charging the separate actor and critic accounts twice. Both the nearest-node neural ablation and the spline receive a strengthened certificate. Their complete matched frontiers isolate certificate sharpness from first-target work and actual policy cost.

The numerical implementation starts from economic primitives. In the scalar investment economy, it fits a fresh ReLU continuation backward against targets using its own previously fitted future network. In the two-capital economy, it trains hidden locations in a convex squared-ReLU continuation, with continuous simplex actions and continuous uniform innovations. The second experiment includes a flexible nonnegative ridge dictionary as well as a quadratic projection. Each comparator constructs its own future targets and uses the same final economic criterion. Neither conventional values nor inherited neural weights are used as labels for the neural services.

We report the complete 150-service catalogue, not only successful candidates. In the original capped scalar services the neural procedure reaches the all-state target 0.06 in all twelve cases and 0.04 in six, whereas the spline reaches both targets in every case. In the coupled services the neural procedure reaches 0.25 in six cases and 0.10 in five; the flexible ridge reaches both in all six. These statements describe the original service, including its original verification cap.

A subsequent diagnostic holds both the neural weights and the deployed policies fixed. Refining only residual verification certifies all six previously unresolved scalar policies at 0.04 and the previously unresolved coupled policy at 0.10. The original failures are not relabeled. The diagnostic shows that they do not, by themselves, identify a failure to construct an accurate neural policy. It does not identify the population success probability of the optimizer, establish a new end-to-end runtime, or change any policy's cost.

The comparative findings are equally integral to the paper. The original scalar signed policy comparisons frequently favor the spline. The new coupled direct intervals all contain zero and therefore do not order the policies. The precision controller genuinely escalates after rejected low-precision proposals, but its complete recorded work does not improve on fixed binary64 in this experiment. These findings distinguish the certified capability of the construction from a comparative advantage that would require additional evidence.

The paper retains its original economic subject and full theory. The exposition now follows the economic problem, the operator, the policy theorem, and the evidence. The complete previous article, proof supplement, and economic applications are materialized unchanged within the current source tree as preserved companions; the preservation map records their exact identities. No historical theorem or application is deleted to obtain a shorter active exposition. The present article and new technical supplement state and prove the additional results used by the new evidence.

1.1. *Relation to the literature*

Neural Bellman Operators sit within approximate dynamic programming, fitted-value methods, and policy iteration, not outside them. Regression-based continuation methods and simulation-based dynamic programming supply important conventional benchmarks (Judd, 1998, Powell, 2011, Longstaff and Schwartz, 2001, Ernst et al., 2005). Actor-critic and approximate policy-iteration methods separate evaluation and improvement (Konda and Tsitsiklis, 1999, Scherrer et al., 2015); safe-improvement arguments likewise distinguish a proposed action from a verified

gain (Kakade and Langford, 2002, Pirodda et al., 2013). We use these distinctions to organize the economic object returned by the algorithm.

Neural methods for differential equations provide ways to approximate values and their derivatives (Sirignano and Spiliopoulos, 2018, Han et al., 2018, Hornik et al., 1990). Such approximation does not on its own verify a returned dynamic policy or its numerical execution. Our certificates can be applied to neural or conventional continuations. The neural-specific construction lies in fresh own-future training and, in the coupled experiment, learned hidden locations. Its comparative value must be isolated against flexible conventional methods, not inferred from the certificate’s validity.

The general economic formulation admits recursive utility (Epstein and Zin, 1989, Duffie and Epstein, 1992). The compact-domain policy theorem uses monotonicity and cash invariance, while the original Gaussian entropic construction has separate moment and spectral conditions. These are different bridges, not interchangeable assumptions. The direct paired-cost identity below applies to conditional expectation; it is not asserted for a nonlinear certainty equivalent. The confidence step uses the empirical Bernstein inequality of Maurer and Pontil (2009). Our additional ingredients are the policy-specific telescoping score, its certificate-derived support, and the numerical enclosure of the original continuous-law comparison.

2. THE CONTROLLED ECONOMY AND THE OPERATOR

2.1. Policy-specific continuation

The original continuous-time economy has an admissible control a and state

$$dS_t = b(t, S_t, a_t) dt + \sigma(t, S_t, a_t) dW_t. \quad (1)$$

The feasible correspondence, invariant or stopped domain, and exit and terminal rewards are economic primitives. For recursive evaluation, the continuation of a fixed policy is represented, on its specified horizon, by

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v.$$

The value is $V^a(t, s) = Y_t^a$ under the well-posedness and comparison hypotheses in the retained controlled-economy analysis. In particular, the recursive argument is the policy's own continuation. Replacing it by the optimal continuation would change the policy-evaluation problem.

The applications companion retains the full boundary specifications, infinite-horizon transversality conditions, utility-domain restrictions, temporal-self equations, and game conditions. A finite-domain residual calculation does not establish any of these by itself. In a game, certifying a specified deviation problem is not the same as establishing the equilibrium fixed point. In continuous time, the controlled generator contains state derivatives; freezing a critic during an actor update does not suppress legitimate derivatives of the represented state-dependent function.

For the constructive implementation and the main policy theorem, consider dates $t = 0, \dots, T$, compact state space X , and feasible correspondence $A_t(x)$. States evolve as

$$X_{t+1} = F_t(X_t, a_t, Z_{t+1}). \quad (2)$$

Let c_t be cost, g terminal cost, $\beta_t > 0$, $B_0 = 1$, and $B_t = \prod_{j < t} \beta_j$. We use a cost-minimization convention in the discrete implementation. It is equivalent to changing signs in an additive reward formulation, but does not replace the original recursive-utility assumptions by a sign convention.

Write

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z_{t+1}))), \quad (3)$$

$$\mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a), \quad V_T^* = g.$$

The conditional evaluations ρ_t are monotone and cash invariant on a domain containing every value and comparison function used below. Feasible measurable selectors exist; compactness, continuity of primitives, and continuity of the evaluated integrals are sufficient conditions in the experiments. The Bellman recursion is valid against all admissible adapted policies. For expectation, J^π is expected discounted cost. For a recursive evaluation it denotes the value generated by that evaluation's own recursion.

2.2. *An executable NBO service*

An implemented operator maps economic data and a current policy to

$$(\hat{f}, \hat{\pi}, \mathcal{C}, W) \quad \text{or} \quad (\text{unresolved}, W, \text{diagnostic}),$$

where $\mathcal{C}$ is an economic certificate and W records the work used. Its stages are target construction, fitting, feasible action selection, independent certification, and durable storage. Critic fitting freezes its future inputs; action selection freezes the resulting critic. In the fresh backward construction, date t uses the already fitted $\hat{f}_{t+1}$, not an externally solved optimal continuation. The terminal target is the primitive g .

A finite optimizer schedule is an algorithm, not a convergence theorem. A service that has not met its target at its cap records failure, with the complete cost of its attempts. The retained factor-training analysis provides a finite constructive theorem for a separate structured class. The nonlinear schedules below instead have an explicit, auditable capped-service estimand.

For a fixed catalogue $\mathcal{K}$ and target ε , define

$$H_m(k, \varepsilon) = \mathbf{1}\{\text{method } m \text{ stores a valid certificate at } \varepsilon \text{ within its cap on object } k\}.$$

(4)

The empirical object is the full array of H_m , costs to the first qualifying checkpoint, and capped costs for failures. Its arithmetic average is a catalogue fraction. A probability over random initializations would require a specified law and a sampling design for that different estimand. A post-review refinement of a failed object's verifier is a new diagnostic and does not change (4) for the original service.

3. CONTINUATION ACCURACY, TRANSPORT, AND POLICY LOSS

3.1. *Centered errors and economic reuse*

Consider a current objective $r_\theta(s, a) + S_\theta(T_\theta(s, a))$ to be maximized. Let $e = \hat{S}_0 - S_0$ be anchor approximation error and $\Delta_\theta = S_\theta - S_0$ the change in the actual future continuation. Suppose the implemented action is δ -optimal for $r_\theta + \hat{S}_0 \circ T_\theta$.

PROPOSITION 1—Centered decision and transport bounds: *The current decision loss satisfies*

$$L_\theta(s, \hat{a}) \leq \delta + \text{osc}_{A_\theta(s)}(e \circ T_\theta) + \text{osc}_{A_\theta(s)}(\Delta_\theta \circ T_\theta). \quad (5)$$

For a common finite action menu with positive weights p_j , set $R_p = \sum_j p_j(e_j - \sum_i p_i e_i)^2$ and $C_p = \max_{i \neq j}(p_i^{-1} + p_j^{-1})$. If $|\Delta_\theta| \leq b_\theta$ on the queried postdecision set, then $L_\theta \leq \delta + \sqrt{C_p R_p} + 2b_\theta$.

If common-innovation future transitions have Lipschitz constants L_k and perturbation bounds $\epsilon_{\theta,k}$, and future rewards have Lipschitz constants H_k and perturbations $\eta_{\theta,k}$ on an invariant domain, put $d_0 = 0$, $d_{k+1} = L_k d_k + \epsilon_{\theta,k}$. For nonnegative reward weights w_k ,

$$|S_\theta - S_0| \leq b_\theta := \sum_k w_k(\eta_{\theta,k} + H_k d_k). \quad (6)$$

The proof compares the true and fitted objective at the two selected actions and couples the future states. The weighted finite-menu bound is Cauchy–Schwarz applied to $e_i - e_j$. Details are in the technical supplement. An additive continuation error common to all actions is irrelevant to the decision. A finite menu, however, does not certify a continuous action set merely because it includes both observed actions.

For recursive maps with sup-norm Lipschitz constants β_k on a common invariant utility domain, a one-step perturbation v_k propagates as $b_k \leq v_k + \beta_k b_{k+1}$. This is the recursive counterpart of (6). Its use requires the common domain; it cannot cross a singular utility boundary by extending a local constant. These transport bounds retain the original refresh logic: use an anchor while its economic allowance covers approximation and future change, and recertify or recompute when that condition is exhausted.

3.2. A direct all-state policy theorem

For continuous fitted functions f_t and a feasible measurable policy π , suppose the verifier supplies, uniformly in x ,

$$\ell_t \leq f_t(x) - \mathcal{T}_t f_{t+1}(x) \leq u_t, \quad t < T, \quad (7)$$

$$Q_t(f_{t+1}; x, \pi_t(x)) - \mathcal{T}_t f_{t+1}(x) \leq \eta_t,$$

$$\ell_T \leq f_T(x) - g(x) \leq u_T.$$

These are enclosures for the actual coefficients and actual returned actions. Neither an optimizer's success flag nor a residual on sampled states replaces them.

THEOREM 2—Centered full-policy certificate: *Under the conditions of Section 2 and (7), every initial state satisfies*

$$0 \leq J_0^\pi(x) - V_0^*(x) \leq G(f, \pi) := \sum_{t < T} B_t(u_t - \ell_t + \eta_t) + B_T(u_T - \ell_T). \quad (8)$$

More specifically,

$$V_0^*(x) \geq f_0(x) - \sum_{t < T} B_t u_t - B_T u_T, \quad (9)$$

$$J_0^\pi(x) \leq f_0(x) + \sum_{t < T} B_t(\eta_t - \ell_t) - B_T \ell_T.$$

Replacing f_t by $f_t + d_t$, with date-specific constants d_t , leaves the bound unchanged after the residual endpoints are shifted by $d_t - \beta_t d_{t+1}$ and the terminal endpoints by d_T .

PROOF: Monotonicity and cash invariance imply $\mathcal{T}_t(v + c) = \mathcal{T}_t v + \beta_t c$ and the analogous identity for the fixed-policy map. Starting from $g \geq f_T - u_T$, backward induction gives the first inequality in (9). Starting from $g \leq f_T - \ell_T$ and using $Q_t(f_{t+1}; x, \pi_t(x)) \leq f_t(x) - \ell_t + \eta_t$ gives the second. Subtraction proves the upper bound; optimality proves the lower bound. The shifted residuals have unchanged widths and unchanged action gaps. *Q.E.D.*

The theorem certifies a full adapted policy, not a finite action menu or a grid of initial states. With expected cost its proof is a Bellman telescoping argument; for a monotone cash-invariant recursive evaluation it is an order argument. The numerical experiment is not needed for its validity. Its numerical usefulness depends on producing small, valid enclosures.

COROLLARY 3—Retained-policy recertification: *Fix the functions f_t , the deployed policy π , valid actor allowances η_t^{old} , and a terminal enclosure. Let a second verifier return*

valid residual enclosures $[\ell_t^{\text{new}}, u_t^{\text{new}}]$ for those same functions. Then

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t^{\text{new}} - \ell_t^{\text{new}} + \eta_t^{\text{old}}) + B_T(u_T^{\text{old}} - \ell_T^{\text{old}}). \quad (10)$$

No retraining or change in deployed actions is required. If desired, intersecting two valid residual enclosures makes the new bound no larger than the old one.

The original actor allowance is valid on the entire state domain, so a finer critic verification mesh does not invalidate it. Conversely, replacing it by an allowance for a newly optimized fine-grid actor would certify a different policy. The implementation records the original actor hash and explicitly discards any fine-grid action proposals generated internally by the verifier. Those proposals' work and transient storage are still charged.

The corollary does not promise that arbitrary precision is attainable by mesh refinement. Nodal residual variation, the retained actor allowance, and the terminal enclosure can leave a positive floor. Nor is the corollary a claim that additional verification improves welfare. It distinguishes the accuracy of a fixed economic decision rule from the sharpness of our knowledge of that accuracy.

3.3. Construction beyond post-training verification

The original trainable-factor construction remains a constructive part of NBO. For a quadratic own-policy continuation $x'P_t x + c_t$, a square-activation network $\|W_t x\|^2/d + \hat{c}_t$ trains every entry of W_t . Its population derivative loss and gradient are

$$\mathcal{L}_t(W; P) = \frac{1}{4} \|W'W - dP\|_F^2, \quad \nabla_W \mathcal{L}_t = W(W'W - dP).$$

If $\sigma_{\min}(W) \geq s > 0$, then $\|W'W/d - P\|_F \leq \|\nabla_W \mathcal{L}_t\|_F/(ds)$. Together with the full-vector actor defect, this gives a policy-error account for a jointly trained hidden factor. The own-policy coefficient is re-evaluated for the returned policy, not silently borrowed from an earlier training policy.

The primitive-budget analysis in the retained article chooses a feasibility witness, coefficient envelopes, rank margins, and update budgets before training. In the Gaussian entropic extension it also verifies positive exponential-moment margins.

Under its stated spectral hypotheses, a scaled orthogonal initialization and bounded gradient steps give the finite training construction proved there. The finite matrix and Gaussian formulas are not a nonlinear dimension-scaling experiment. They also do not imply that Adam or L-BFGS has a deterministic residual cap for a general squared-ReLU network. We preserve both results and keep the two logical statements separate.

4. FROM ECONOMIC PRIMITIVES TO NONLINEAR CERTIFICATES

4.1. Backward fitting and selected policies

For each method, initialize a fresh terminal fit from g . At date t , construct targets from $\mathcal{T}_t f_{t+1}$ using that method's own already fitted future. Fit the current continuation under the frozen schedule. Then propose feasible actions, enclose the Bellman infimum and the selected-action gap, and validate residual coverage. Store a checkpoint before reporting its first target crossing. Continue through the declared ladder only as specified by the original study. The record includes all attempted resolutions, unsuccessful target queries, target-generation work, and storage operations.

The scalar network has 63 ReLU features. Its training uses 513 states, a 257-point target action construction, 400 Adam steps at learning rate 5×10^{-5} , and a final output fit. The scalar verifier uses its own finer action and state ladders. The conventional spline is reconstructed from the same primitives with its own ladder and targets. A conventional target solver is not used to label a neural continuation.

The coupled neural continuation is

$$f_t(x) = a_t + b'_t x + \sum_{j=1}^{48} c_{tj} (w'_{tj} x + v_{tj})_+^2, \quad c_{tj} \geq 0. \quad (11)$$

Hidden locations and weights are trained. Nonnegative coefficient initialization and a finite L-BFGS schedule of 300 iterations act on fresh own-future targets. The ridge comparator instead fixes a 96-feature squared-hinge dictionary and fits nonnegative coefficients to its own targets; zero coefficients may be pruned. Its dictionary seed is fixed by date, so repeated training-seed labels do not create independent random

ridge representations. A quadratic projection is retained as a restricted comparator, not treated as the strongest conventional alternative.

4.2. All-state and continuous-action verification

At each state node, outward arithmetic encloses the actual network value and the action objective. In the coupled economy, convexity in actions and a positive strong-convexity constant yield a continuous-simplex primal–dual gap. A projected optimizer merely proposes the action; its stopping flag is not the certificate. State interpolation then extends the signed residual bounds off-grid. The nearest-node deployed actor receives a separate, uniform allowance for using its node action at the actual state.

Uniform innovations are integrated through positive-part moments, rather than replaced by a sampled innovation law. The elementary identity for a uniform variable U on $[-h, h]$ is

$$\mathbb{E}(y + U)_+^2 = \frac{(y + h)_+^3 - (y - h)_+^3}{6h}, \quad h > 0.$$

Successive integration gives the corresponding formula for independent coordinates and affine neuron arguments; zero coefficients are handled by the lower-dimensional formula. Interval evaluation encloses cancellation and rounding. The supplement specifies the mesh and action bounds used in the inherited verifier.

The residual account distinguishes a nodal enclosure from state coverage, actor error, and terminal error. A nodal enclosure contains the effects of representation, fitting, Bellman optimization, and arithmetic; it is not an identified decomposition into a pure approximation error and a pure optimizer error. In particular, subtracting a training loss from the residual does not estimate either component. The diagnostic reports every separately available allowance without counting nested subcomponents twice.

4.3. *Work, dimension, and execution*

For tensor state, action, and innovation covers of sizes K_x, K_a, m , a direct verifier pays work of order

$$\sum_{t < T} K_x K_a \{m(C_F + P_{t+1}) + C_\rho\} + \sum_{t \leq T} K_x P_t + W_{\text{storage}}, \quad (12)$$

where P_t counts network evaluation work. Actor storage is of order $TK_x d_a$. Analytic integration replaces m by the work of the moment formula, not by zero. With tensor covers, K_x scales as $h_x^{-d_x}$; analogous factors arise for action and innovation discretizations. A dimension-explicit certificate is therefore not a dimension-free algorithm.

If a Lipschitz innovation approximation contributes at most ζ_t to Q_t , widening residual endpoints by ζ_t and the actor allowance by $2\zeta_t$ adds $4 \sum_t B_t \zeta_t$ to (8). This is a discretization account, not permission to replace a continuous law without charging its error. Independent policy queries may have their own scenario-tree or simulation cost and are recorded separately.

The complete service includes own-future targets, fitting, unsuccessful refinement, action extraction, verification, applicable policy queries, serialization, and durable checkpoint storage. Initialization and outer-process time are retained separately. New local diagnostic clocks are not added to old GitHub-hosted clocks to manufacture an end-to-end estimate. All reported arithmetic certificates concern stored binary coefficients with outward enclosures. The inherited execution account treats rounded state acquisition and transition/cost evaluation separately from ideal policy loss.

5. DIRECT COMPARISONS OF ECONOMIC POLICY COSTS

A smaller upper bound on $J^\pi - V^*$ need not mean a smaller J^π . We therefore compare the neural and ridge policies directly at common states, using common innovations and the expectation version of (3).

For method m , let X^m follow its own fixed policy and define

$$Z_m = f_0^m(x_0) + \sum_{t < T} B_t \{Q_t(f_{t+1}^m; X_t^m, \pi_t^m(X_t^m)) - f_t^m(X_t^m)\} + B_T \{g(X_T^m) - f_T^m(X_T^m)\}. \quad (13)$$

This is a control-variate score for actual policy cost, not the difference of two regret bounds.

THEOREM 4—Paired economic score and support: *For fixed policies and integrable continuations under conditional expectation, $\mathbb{E}Z_m = J_0^{\pi^m}(x_0)$. If method m satisfies (7), then every trajectory satisfies*

$$L_m \leq Z_m \leq U_m, \quad (14)$$

$$L_m = f_0^m(x_0) - \sum_{t < T} B_t u_t^m - B_T u_T^m,$$

$$U_m = f_0^m(x_0) + \sum_{t < T} B_t (\eta_t^m - \ell_t^m) - B_T \ell_T^m.$$

Consequently $Z_N - Z_R$ has expectation $J^{\pi^N} - J^{\pi^R}$ and support $[L_N - U_R, U_N - L_R]$, whose width is $G_N + G_R$. These statements hold under any coupling of the two methods that preserves each policy's innovation law.

PROOF: Conditional expectation gives $\mathbb{E}Q_t(f_{t+1}^m; X_t^m, \pi_t^m) = \mathbb{E}c_t + \beta_t \mathbb{E}f_{t+1}^m(X_{t+1}^m)$. Summing telescopes all continuation terms. The selected-action gap is between zero and η_t^m , so each centered summand in (13) is between $-u_t^m$ and $\eta_t^m - \ell_t^m$. The terminal summand is between $-u_T^m$ and $-\ell_T^m$. Summing proves the support statement and its width. *Q.E.D.*

To evaluate the original continuous law, simulate uniform innovation bins and enclose all within-bin innovations. Interval propagation includes every actor that can be selected when a state enclosure crosses a nearest-node boundary. Suppose the resulting endpoint pair satisfies $A_i \leq Z_{N,i} - Z_{R,i} \leq D_i$ and lies in a known support interval of width B . For independent draws of bins, the lower and upper endpoints are bounded independent observations. Let $\bar{A}, \bar{D}$ and s_A^2, s_D^2 be their sample means

and variances. For q predeclared comparisons, set

$$r(s^2) = \sqrt{\frac{2s^2 \log(4q/\alpha)}{n}} + \frac{7B \log(4q/\alpha)}{3(n-1)}. \quad (15)$$

Applying the one-sided empirical Bernstein inequality to the lower and upper endpoints separately, then taking a union bound, gives simultaneous coverage at least $1 - \alpha$ for

$$[\bar{A} - r(s_A^2), \bar{D} + r(s_D^2)] \quad (16)$$

over all q comparisons. Mean, variance, logarithm, and square-root calculations are outward enclosed in the implementation; intersecting with the deterministic support preserves coverage. The proof and numerical details are in the supplement.

The coverage statement is conditional on frozen training outputs and an independent simulation model. Fixed PCG64 streams make this implementation reproducible; a fixed stream is not itself a proof of probabilistic independence. No policy is selected using these simulations. Intervals crossing zero are reported as unresolved, not as equivalence or as evidence of a superior neural policy.

6. INVESTMENT ECONOMIES AND THE COMPLETE EVIDENCE

6.1. Design and provenance

The original science design has twelve scalar neural services, twelve scalar spline services, six coupled neural services, six coupled ridge services, six coupled quadratic services, and 36 services for each of three precision rules. The successful source-bound execution contains all 150 records. The original first run, with four nonfinite-network exceptions, is retained. Its numerical inverse-softplus initialization was subsequently replaced by an algebraically equivalent stable expression; economic primitives, seeds, targets, budgets, and comparators were unchanged. We do not treat the failed first run as nonexistent or combine its clocks with the successful run.

The R44 protocol was committed before the additional diagnostics were executed. It is a post-review protocol, not an original preregistration. It selects the seven unresolved tight-target neural objects by their already observed outcomes and

freezes their weights and policies. The paired comparison uses all six coupled pairs and all four originally declared states. A source manifest binds the original inputs, the diagnostic code, outcomes, and publication sources. Every original record and all five original science-source hashes were replayed.

6.2. Scalar and coupled primitives

The scalar experiment retains the original bounded investment state, price and risk cells, three declared training seeds, continuous feasible action interval, and the independent own-policy evaluator. Prices are one and four. Both methods construct their own continuations from the same primitive problem. The retained evidence and code specify every scalar primitive; the full economic derivation remains in the original investment section and applications.

In the coupled economy, $x \in [0, 1]^2$, $T = 4$, $\beta = 15/16$, and $A = \{a \geq 0 : a_1 + a_2 \leq 1/8\}$. With independent $z_j \sim U[-1/32, 1/32]$,

$$F_j(x, a, z) = \frac{1}{8} + \frac{1}{2}x_j + \frac{1}{8}x_{3-j} + \frac{1}{16}x_j(1 - x_{3-j}) + a_j + z_j.$$

This transition maps the stated state-action-innovation domain into the state domain. Define

$$s(x) = \sum_{j=1}^2 (x_j - 11/16)^2 + 2(1/2 - x_1 - x_2)_+^2 + \frac{1}{4}(x_1 - x_2)^4.$$

The stage cost is $s(x) + p \sum_j a_j^2 + \frac{1}{2}a_1a_2 + 4 \sum_j a_j^4$; terminal cost doubles the squared-target term in s and retains its remaining terms. Prices $p \in \{1, 4\}$ penalize costly investment. The shared simplex represents a common resource constraint, the transition couples the two capital stocks, and the state cost penalizes aggregate shortage and imbalance. This is a stylized controlled economy, not an empirically calibrated welfare exercise.

For both coupled methods the final all-state verification ladder is $N = 16, 32, 64, 128$. The action objective has strong-convexity lower bound $2p - 1/2 > 0$. Uniform positive-part moments integrate continuous uncertainty analytically. The finest original actor has $4(129)^2 \cdot 2 = 133,128$ stored scalar actions. This actor cost is distinct from continuation-network storage.

6.3. *Original attainment and comparative outcomes*

TABLE I

ORIGINAL CAPPED-SERVICE TARGET ATTAINMENT

Method	All-state target	Certified
Scalar neural	0.06	12/12
Scalar neural	0.04	6/12
Scalar spline	0.06	12/12
Scalar spline	0.04	12/12
Coupled neural	0.25	6/6
Coupled neural	0.10	5/6
Coupled ridge	0.25	6/6
Coupled ridge	0.10	6/6
Coupled quadratic	0.25	0/6
Coupled quadratic	0.10	0/6

Table I reports the original capped-service outcomes. The original scalar neural bound is larger than its matched spline bound in every one of the twelve pairs. At the first common 0.06 crossing, 29 of 48 signed neural-minus-spline intervals have positive lower endpoints, none has a negative upper endpoint, and 19 contain zero. At 0.04, the six common pairs give 17 positive intervals, none negative, and seven overlaps. All lie within the prespecified descriptive margin ± 0.005 . This establishes numerical closeness at those states but is not an equivalence design; where signs are identified, they favor the spline.

The coupled ridge reaches both targets in all six services. It has the smaller final gap in four pairs and the neural method in two. At 0.25 the ridge reaches the target faster in all six original observations. At 0.10 it is faster in four of the five pairs in which both pass, while the neural method is faster in one and fails in the remaining pair. These singleton nonlinear timing observations are descriptive. The ridge continuation has storage comparable to the neural continuation, so its strength cannot be dismissed as a comparison with an artificially restricted quadratic.

6.4. Diagnosing tight-target failures without changing policies

TABLE II
ADDITIONAL VERIFICATION OF UNCHANGED NEURAL POLICIES

Service	Original bound	Refined bound	Target	Attained
041	0.04092128	0.03068472	0.04	Yes
089	0.04118465	0.03090917	0.04	Yes
120	0.04112077	0.03085353	0.04	Yes
124	0.04093160	0.03070697	0.04	Yes
128	0.15223570	0.09775755	0.10	Yes
145	0.04136567	0.03108538	0.04	Yes
147	0.04101560	0.03077643	0.04	Yes

For the six unresolved scalar policies, only the residual state grid is refined from $N = 2048$ to $N = 4096$, retaining the original action verification resolution $A = 1024$. For the coupled policy, residual verification uses $N = 256$ instead of 128. The original actor allowances and terminal enclosures remain in (10); all original action arrays and networks are hash-checked unchanged. Fine-grid actor proposals generated internally are discarded, not deployed.

Table II gives all seven results. The retained scalar policy bounds become approximately 0.03068–0.03109, below 0.04. The retained coupled policy bound becomes 0.09775756, below 0.10. There are zero training updates and zero newly stored deployed-action scalars. The coupled diagnostic still pays for a fine verification mesh with 66,049 states, including transient fine-grid proposals; retaining the original actor does not make verification free.

The original coupled decomposition is informative. Its discounted state-cover contribution to nonterminal residual widths is about 0.07268224, its actor allowance is 0.03287249, its remaining nodal residual enclosure is 0.04574738, and its terminal contribution is 0.00093360. These sum to the original bound up to displayed rounding. The extra mesh quarters the principal second-order coverage term while retaining the actor and terminal terms. The resulting crossing does not establish

that representation and optimization errors are separately small; it establishes the required full-policy bound for the unchanged policy.

The diagnostic answers a specific method question. The original capped service was less reliable at its tight target, but the saved neural objects were not necessarily inadequate policies. All seven selected policies admit the tighter certificates when verification is extended under a fixed rule. This finding does not alter the original attainment table or justify a comparison between a selectively refined neural certificate and an unrefined comparator as evidence of a work advantage.

6.5. *Direct coupled economic differences*

TABLE III
DIRECT NEURAL-MINUS-RIDGE COST COMPARISONS, SIMULTANEOUS 99 PERCENT INTERVALS

Price	Comparisons	Positive	Negative	Unresolved
1	12	0	0	12
4	12	0	0	12

For each of the six original neural/ridge pairs, select each method's first 0.25 crossing. Compare at $(1/8, 1/8)$, $(1/4, 1/2)$, $(1/2, 1/4)$, and $(3/4, 3/4)$. Each comparison uses 65,536 common innovation-bin trajectories, 40-bit uniform bins, and the own-policy score in (13). The total family error is 0.01 across 48 one-sided bounds. The complete 24 intervals, endpoint moments, seeds, ambiguity counts, and support bounds appear in the supplement and machine-readable records.

All 24 direct confidence intervals contain zero. Thus the coupled evidence now addresses the actual neural-minus-ridge policy-cost difference, but does not rank the methods. This is stronger information than subtracting regret bounds and different information from a smaller all-state certificate. The simultaneous interpretation pertains to the declared states and fixed policies under the simulation model, not to every state in the compact domain. The all-state policy guarantees remain the separate deterministic statements in Table I and Theorem 2.

6.6. Precision and complete recorded work

TABLE IV
ORIGINAL REPEATED PRECISION CATALOGUE

Method	Certified	Accepted32	Rejected32	Accepted64
Fixed32	6/36	147	30	0
Fixed64	36/36	0	0	474
Adaptive	36/36	348	126	126

The precision experiment repeats twelve dimension/conditioning/target objects three times. Fixed binary64 and adaptive each certify all 36 services; fixed binary32 certifies six. Fixed64 and adaptive each make 474 accepted updates. Adaptive makes 348 accepted binary32 updates, 126 rejected binary32 proposals, and 126 accepted binary64 updates. Every rejection is charged.

Adaptive is slower in 26 of the 36 matched recorded services and faster in ten; aggregate recorded time is about 0.98% larger. The controller's activation therefore demonstrates its numerical mechanism, not a complete-work saving in this implementation. This result does not invalidate the conditional numerical acceptance theorem. It prevents the theorem from being mistaken for an empirical efficiency conclusion.

6.7. What the evidence identifies

The evidence identifies a finite from-primitives nonlinear construction, all-state certificates for its returned policies, a useful retained-policy verification refinement, and direct economic comparisons under the declared state and innovation design. It also identifies the continued strength of conventional comparators. It does not identify a general neural speed advantage, a nonlinear high-dimensional frontier, or a population probability of optimizer success.

An adaptive sparse-grid or adaptive-partition comparator has not been executed for the original neural optimizers. Neither has a nonlinear dimension frontier or repeated isolated timing of every original scalar and coupled service. Section [7.3](#)

adds a separate predeclared horizon/price catalogue with repeated isolated timing for the deterministic constructive backend and spline; it does not replace the original experiments. The retained dimension-explicit analysis exposes the tensor factors those studies would have to confront, rather than treating a two-state experiment as their substitute. The general economic framework and conditional constructive results remain in force; the missing experiments concern comparative evidence for broader implementation claims.

7. A CONSTRUCTIVE NEURAL SERVICE WITH A PRIMITIVE ERROR BUDGET

The policy theorem is deliberately independent of the representation of the continuation. A construction theorem must do more: it must produce a representation, feasible actions, and verified residuals from economic primitives. This section supplies such a theorem for a deterministic ReLU backend. It does not assume convergence of the nonlinear optimizers used in the earlier experiments. Its distinguishing property is that the continuation has a prescribed Lipschitz constant even when its Bellman labels are inexact.

7.1. Primitive conditions and the neural architecture

Consider the compact Bellman economy of Section 2, with a common action set $A \subset \mathbb{R}^a$ and state set $X \subset \mathbb{R}^d$. All distances in this section use the ℓ^1 norm. The innovation law and the monotone, cash-invariant functional ρ_t do not depend on the state or action. The transition remains in X . Suppose known nonnegative constants satisfy, uniformly in the innovation,

$$|c_t(x, a) - c_t(y, b)| \leq C_{x,t} \|x - y\|_1 + C_{a,t} \|a - b\|_1, \quad (17)$$

$$\|F_t(x, a, z) - F_t(y, b, z)\|_1 \leq F_{x,t} \|x - y\|_1 + F_{a,t} \|a - b\|_1. \quad (18)$$

The terminal cost is L_g -Lipschitz. These assumptions are additional conditions on the constructive backend, not a replacement for the more general feasible correspondences in the original economic formulation. A state-dependent feasible correspondence requires its own feasible-cover and transport constants.

We require an effective numerical description of the primitives: certified finite state and action covers can be obtained at every positive radius; their nodes are feasible; and a numerical evaluator can enclose each Bellman query to any requested positive tolerance in finite work. The last condition includes evaluation of ρ_t , not just evaluation of the network. For expectation with a compact innovation domain, known cell probabilities and a certified innovation cover provide one such evaluator. If F_t is $F_{z,t}$ -Lipschitz in the innovation, replacing each cell by a representative at distance at most h_z contributes at most $\beta_t L_{t+1} F_{z,t} h_z$. Arithmetic, probability-mass approximation, and any remaining integration error must be added. The experiment below instead integrates a piecewise-linear network analytically under the original continuous uniform law.

Set

$$L_T = L_g, \quad L_t = C_{x,t} + \beta_t F_{x,t} L_{t+1}, \quad L_t^a = C_{a,t} + \beta_t F_{a,t} L_{t+1}. \quad (19)$$

Choose state nodes $x_{t,i}$ of covering radius h_t and action nodes $a_{t,j}$ of covering radius k_t . Given the continuation already constructed at date $t + 1$, compute numbers $\hat{q}_{t,ij}$ satisfying

$$|\hat{q}_{t,ij} - Q_t(f_{t+1}; x_{t,i}, a_{t,j})| \leq e_t. \quad (20)$$

Define $y_{t,i} = \min_j \hat{q}_{t,ij}$, choose a minimizing action index $j(t, i)$ with a fixed tie rule, and construct

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (21)$$

At date T , use labels within e_T of g and the same formula with constant L_g . At any other date, deploy the action $a_{t,j(t,i(x))}$ attached to a nearest state node, again with a fixed tie rule. The action is feasible by construction. No future optimal value or conventional comparator is used as a label.

Formula (21) is an exact ReLU construction, not an appeal to a universal approximation theorem. Absolute values use $|u| = \max\{u, 0\} + \max\{-u, 0\}$; minima use $\min\{u, v\} = u - \max\{u - v, 0\}$. A balanced tree of these gates implements the finite minimum. With K_t state nodes, the resulting sparse affine-ReLU circuit has $O(dK_t)$ coefficients and $O(1 + \log K_t)$ depth with unrestricted affine fan-in. Signed interme-

1 diate values can be carried by their positive and negative parts. The construction is 1
 2 related to the classical Lipschitz extension formula of [McShane \(1934\)](#); neither that 2
 3 formula nor its elementary ReLU realization is claimed as new. Here its role is to 3
 4 close an own-future dynamic construction, including the deployed actor and every 4
 5 numerical allowance. 5

6
 7 **THEOREM 5**—Constructive own-future neural policy: *Under the preceding condi-* 7
 8 *tions, the network (21) is L_t -Lipschitz at every date, independently of the numerical label* 8
 9 *errors. Its residual and its deployed actor satisfy* 9

$$-e_t \leq f_t - T_t f_{t+1} \leq L_t^a k_t + e_t + 2L_t h_t, \quad (22)$$

$$Q_t(f_{t+1}; x, \pi_t(x)) - T_t f_{t+1}(x) \leq L_t^a k_t + 2e_t + 2L_t h_t. \quad (23)$$

10
 11
 12
 13 *At the terminal date, $-e_T \leq f_T - g \leq e_T + 2L_g h_T$. Consequently the cost of the returned* 13
 14 *policy, relative to the optimum over all admissible adapted policies, is uniformly bounded by* 14
 15

$$\mathcal{G}_{\text{con}} = \sum_{t < T} B_t \{2L_t^a k_t + 4e_t + 4L_t h_t\} + B_T \{2e_T + 2L_g h_T\}. \quad (24)$$

16
 17
 18 *For fixed nodes and L_t , changing every label by at most δ changes the represented function* 18
 19 *by at most δ in the uniform norm; it does not change the Lipschitz bound.* 19
 20

21 **PROOF:** Monotonicity and cash invariance imply nonexpansiveness in the 21
 22 uniform norm. Thus $Q_t(f_{t+1}; \cdot, a)$ and $T_t f_{t+1}$ are L_t -Lipschitz, while the action 22
 23 modulus is L_t^a . At a node, minimization of the numerical queries gives $-e_t \leq$ 23
 24 $y_{t,i} - T_t f_{t+1}(x_{t,i}) \leq L_t^a k_t + e_t$. Each cone in (21) is L_t -Lipschitz, as is their finite min- 24
 25 imum. The lower residual bound follows from the triangle inequality for every 25
 26 cone; the upper bound uses a nearest node twice, once for the cone and once 26
 27 for the Bellman value. At that node the selected action is within $L_t^a k_t + 2e_t$ of the 27
 28 continuous-action infimum. Transport to x costs at most $2L_t h_t$. The terminal argu- 28
 29 ment is identical without action minimization. Substitution of these widths and 29
 30 actor allowances into Theorem 2 gives (24). Finally, minima of two arrays whose 30
 31 entries differ by at most δ differ by at most δ . The supplement gives the construction, 31
 32 measurability, and work details. Q.E.D. 32

COROLLARY 6—Deterministic termination at an economic tolerance: *Let $\varepsilon > 0$, $H = \sum_{t=0}^T B_t$, and $s = \varepsilon/H$. At nonterminal dates choose*

$$e_t \leq s/8, \quad L_t^a k_t \leq s/8, \quad L_t h_t \leq s/16. \quad (25)$$

At the terminal date choose $e_T \leq s/4$ and $L_g h_T \leq s/4$. Terms with zero coefficient impose no restriction. Then the finite constructive service returns a feasible neural policy with loss at most ε . No probability law over initializations and no optimizer stopping criterion is needed.

The result follows because each discounted bracket in (24) is at most s . Effective compactness and the numerical evaluator give finite work for these choices. Arbitrarily small tolerances may require arbitrary precision; the corollary does not promise termination in fixed binary64. A finite implementation ladder can still return a recorded failure, even though the unbounded constructive algorithm terminates under the stated hypotheses.

The earlier scalar constructive result in the retained paper implemented piecewise-linear interpolation as a ReLU network. Its sufficient allocation controlled a label-error-to-mesh ratio in the propagated slope. Formula (21) works in arbitrary finite state dimension and enforces the slope bound directly. In particular, label and action allowances can be allocated at order ε without a separate slope-preservation requirement that those allowances be quadratic in the state spacing. This is an improvement of that sufficient proof budget, not a lower bound for all spline methods or evidence that conventional Lipschitz projection is unavailable.

7.2. What the construction costs

The construction does not eliminate state coverage. With K_t state nodes, J_t action nodes, M_t innovation representatives, and network evaluation cost P_{t+1} , direct target formation requires

$$\sum_{t < T} K_t J_t \{M_t (C_{F,t} + P_{t+1}) + C_{\rho,t} + C_{c,t}\} + O\left(\sum_{t=0}^T d K_t\right) \quad (26)$$

arithmetic operations under the indicated evaluation model. $C_{\rho,t}$ includes the finite risk functional and probability processing. Certified integration can replace the innovation sum but its own cost must be charged. Bit lengths, adaptive arithmetic, and nearest-node policy queries are additional explicit costs, not constants suppressed by a dimension claim. A rectangular state cover with radius h_t can require order h_t^{-d} nodes; a rectangular action cover can require order k_t^{-a} actions. The theorem is therefore a dimension-explicit construction, not an assertion that a nonlinear high-dimensional frontier has been executed or that the curse of dimensionality has been removed.

In one state dimension, the finite envelope can be compiled exactly into a piecewise-linear representation. Two passes compute the greatest L -Lipschitz minorant of the node labels. Between neighboring nodes the envelope is the minimum of the two adjacent affine cones. At most one additional intersection is needed per interval. Rational knots and values permit exact antiderivatives; an interval evaluator encloses their numerical evaluation. This compilation evaluates the defining network exactly and does not substitute labels from a separately solved Bellman problem. The spline comparator receives the same exact-integration acceleration, so it is not charged for an artificial dense neural evaluation loop.

7.3. A predeclared matched service catalogue

The new experiment tests complete services, not only previously failed neural candidates. It uses a further nonlinear investment laboratory with $x \in [0, 1]$, $a \in [0, 1/4]$, $\beta = 15/16$, and independent z uniform on $[-1/32, 1/32]$:

$$F(x, a, z) = \frac{1}{32} + \frac{11}{16}x + \frac{1}{16}x(1 - x) + a + z, \quad (27)$$

$$c_p(x, a) = (x - \frac{11}{16})^2 + pa^2 + 4a^4 + 2(\frac{3}{8} - x)_+^2, \quad (28)$$

$$g(x) = 2(x - \frac{11}{16})^2 + 2(\frac{3}{8} - x)_+^2. \quad (29)$$

The invariant domain follows from monotonicity in x, a, z and the two extreme transitions 0 and 1. Valid primitive state moduli are $C_x = 23/8$, $F_x = 3/4$, and $L_g = 17/4$; action moduli are $C_a = p/2 + 1/4$ and $F_a = 1$. The downside term penalizes a low capital stock, and the quartic investment cost discourages rapid

adjustment. This is a numerical economic laboratory, not an estimated or calibrated quantitative exercise.

The complete design was fixed before execution: horizons $\{2, 4\}$, prices $\{1, 4\}$, two methods, three isolated repetitions, and the full common state/action ladder $N = A \in \{16, 32, 64, 128, 256, 512\}$. The targets are 0.5, 0.25, and 0.125 in units of discounted cost. These are policy-loss tolerances, not equivalence margins for a neural-minus-spline contrast. Every rung restarts from terminal primitives and constructs its own future backward. The neural method uses (21); the spline uses linear interpolation of its own labels and propagates its actual exact slope bound. Both evaluate the same continuous innovation law analytically and round labels to dyadic multiples of 2^{-40} . All numerical query errors are outwardly enclosed and enter the certificate.

All rungs are retained, including those preceding the first target crossing. The clock to a target includes every preceding rung, target formation, rational compilation, actor construction, verification, and durable checkpoint output. Fresh processes run sequentially on one selected CPU with one numerical-library thread and an identical non-economic warm-up. Imports and warm-up are excluded from the internal clock but included in a separate process clock. CPU frequency is not fixed. Independent policy-cost comparison is not performed in this catalogue and is not assigned a zero cost. Three timing repetitions describe local dispersion, not a population law for neural optimization.

The neural backend certifies all twelve services at 0.5 and 0.25, and nine of twelve at 0.125. The spline certifies all twelve at every target. The three neural failures are repetitions of the same horizon-four, price-four economy: the final gap is approximately 0.12503, just above 0.125. They remain failures rather than being rounded into attainment. Across the 33 common successful method/target/repetition comparisons, the spline reaches the target earlier in every recorded observation. Thus the new execution establishes reproducible complete construction and a transparent frontier, not neural cost superiority. The deterministic termination corollary concerns its specified sufficient budget; the predeclared finite ladder need not contain that budget. The finite catalogue measures the deterministic backend just specified. It neither overwrites the original neural optimizer's attainment array nor turns the

TABLE V

FIRST CERTIFIED STATE / ACTION RESOLUTION IN THE COMPLETE CONSTRUCTIVE CATALOGUE

Generator	T	p	0.5	0.25	0.125	Final gap
ReLU envelope	2	1	64	128	256	0.06132
Spline	2	1	64	128	256	0.05619
ReLU envelope	2	4	128	256	512	0.06274
Spline	2	4	64	128	256	0.05799
ReLU envelope	4	1	128	256	512	0.12236
Spline	4	1	128	256	512	0.09711
ReLU envelope	4	4	256	512	–	0.12503
Spline	4	4	128	256	512	0.10221

Entries are first qualifying $N = A$ on the fixed ladder. Each row represents three identical deterministic policy/certificate arrays, not three independent economic tasks. A dash is a retained failure at the declared cap $N = 512$. Final gaps use the cap for both methods.

TABLE VI

COMPLETE PREFIX WORK AT TARGET 0.125, OR AT THE FAILED CAP

Generator	T, p	N	Queries	Median s	Range s	Exit
ReLU envelope	2,1	256	176,586	0.3009	[0.2982, 0.3094]	target
Spline	2,1	256	176,586	0.2144	[0.2136, 0.2365]	target
ReLU envelope	2,4	512	702,924	0.8450	[0.8354, 0.8472]	target
Spline	2,4	256	176,586	0.2151	[0.2119, 0.2153]	target
ReLU envelope	4,1	512	1,405,848	1.5540	[1.5497, 1.5626]	target
Spline	4,1	512	1,405,848	1.2654	[1.2341, 1.2745]	target
ReLU envelope	4,4	512	1,405,848	1.5495	[1.4602, 1.5506]	cap
Spline	4,4	512	1,405,848	1.2705	[1.2625, 1.2810]	target

Queries sum all Bellman evaluations on earlier failed rungs and the reported rung. Clocks include construction, verification and durable checkpoints. Ranges contain the three isolated observations. A cap row is not time to successful certification. Independent economic comparison is not included.

seven retained-policy diagnostics into prospective observations. The theorem supplies a method-level termination statement for this backend; the recorded services establish only the achieved targets, costs, and storage on the declared economies. A smaller certificate is not a lower realized policy cost. The new numerical comparison and the earlier direct policy-cost comparisons therefore remain distinct throughout.

8. NEURAL CONTINUATION WITH AN ACTION WITNESS

A constructed continuation contains information about the action that produced each Bellman label. Discarding that information and subsequently controlling an unrelated actor can charge the same approximation twice. We now retain the label’s feasible action through the minimum circuit. This changes the returned policy, not merely the description of an old certificate. The construction continues to use the original economic primitives and its own fitted future.

8.1. *A one-sided policy sandwich*

Write $\mathcal{T}_t^\pi f_{t+1}(x) = Q_t(f_{t+1}; x, \pi_t(x))$. Suppose that, uniformly on the invariant domain,

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq d_t, \quad \ell_T \leq f_T - g \leq u_T. \quad (30)$$

The quantities u_t and d_t are signed upper bounds. Their separate absolute values need not enter a policy-loss account.

PROPOSITION 7—One-sided policy sandwich: *Under the monotonicity, cash-invariance and Bellman comparison conditions of Section 2, inequalities (30) imply*

$$0 \leq J_0^\pi(x) - V_0^*(x) \leq \sum_{t < T} B_t(u_t + d_t) + B_T(u_T - \ell_T). \quad (31)$$

The bound is invariant to additive date-specific shifts of the continuation. It holds for the stated recursive evaluations as well as expectation; no simulation identity is required.

PROOF: Backward order comparison gives $V_0^* \geq f_0 - \sum_{t < T} B_t u_t - B_T u_T$ and $J_0^\pi \leq f_0 + \sum_{t < T} B_t d_t - B_T \ell_T$. Subtraction proves the result. A shift $f_t \mapsto f_t + b_t$ adds $b_t - \beta_t b_{t+1}$ to u_t and subtracts it from d_t , leaving their sum unchanged. *Q.E.D.*

The centered theorem remains valid: its choice $d_t = \eta_t - \ell_t$ recovers (8). The new proposition avoids that indirect bound when the selected-policy residual can be certified jointly with the continuation. It is an order comparison, not a new claim of neural superiority.

8.2. Retaining the minimizing cone's action

Use the primitive conditions, grids, query tolerances and labels of Section 7. Let $a_{t,i}$ denote the action whose numerical query attains $y_{t,i}$. In place of the nearest-node actor, return

$$i_t(x) \in \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad \pi_t^w(x) = a_{t,i_t(x)}. \quad (32)$$

A fixed tie convention makes this finite selector measurable. The common action set makes each selected action feasible, even when the attaining cone is not centered at a nearest node. The critic remains the exact affine-ReLU circuit. The actor additionally uses the circuit's comparison index; a discontinuous selector is not asserted to be a continuous ReLU output.

THEOREM 8—Witness-preserving neural construction: *Under the conditions of Theorem 5, the policy (32) satisfies*

$$Q_t(f_{t+1}; x, \pi_t^w(x)) - f_t(x) \leq e_t. \quad (33)$$

Consequently its all-state loss relative to the optimum over all admissible adapted policies is at most

$$\mathcal{G}_w = \sum_{t < T} B_t \{L_t^a k_t + 2e_t + 2L_t h_t\} + B_T \{2e_T + 2L_g h_T\}. \quad (34)$$

The nonterminal allowance is half the separated sufficient allowance in (24); the terminal account is unchanged. This comparison of certified bounds does not assert that the two policies have ordered realized costs.

PROOF: For the attaining index $i = i_t(x)$, the numerical label and its action obey $Q_t(f_{t+1}; x_{t,i}, a_{t,i}) \leq y_{t,i} + e_t$. Transporting this same action to x costs at most $L_t \|x - x_{t,i}\|_1$. Hence its objective is at most $f_t(x) + e_t$. The unchanged optimal-residual upper bound is $u_t = L_t^a k_t + e_t + 2L_t h_t$. Apply Proposition 7 with $d_t = e_t$ and the unchanged terminal enclosure. Q.E.D.

The label, the cone and the selected action are one mathematical object in this argument. Retaining only a Lipschitz-regularized label value without an attaining

original action index would not justify (33). The supplement proves that the scalar compiler preserves precisely this information, including dominated labels, rational intersections and ties.

COROLLARY 9—Resource and execution allowances: *Let $s = \varepsilon / \sum_{t=0}^T B_t$. At non-terminal dates it suffices to choose*

$$e_t \leq s/4, \quad L_t^a k_t \leq s/4, \quad L_t h_t \leq s/8,$$

and retain the terminal allocation of Corollary 6. Under the same effective-computability hypotheses the construction terminates with loss at most ε .

If the chosen cone exceeds the minimum by at most v_t , state acquisition error is at most r_t in the stated norm, and a feasible executed action is within α_t of that cone's action, replace e_t in (33) by

$$e_t + v_t + 2L_t r_t + L_t^a \alpha_t.$$

Add the discounted execution terms to (34).

The relaxed allocation is a sufficient-budget improvement relative to the earlier proof, not a complexity lower bound for conventional methods. A tensor cover still incurs its dimension factors. Exact state acquisition is the reference experiment; unchecked lookup using rounded switchpoints does not inherit an exact-index guarantee.

8.3. A strengthened conventional comparison

For either the original nearest-node cone actor or the spline actor, let C_i be the node's closed nearest-neighbor cell and let f_t be that method's own continuation. The same selected-label argument gives the computable bound

$$d_t = e_t + D_t, \quad D_t = \max_i \sup_{x \in C_i} \{y_{t,i} + L_t \|x - x_{t,i}\|_1 - f_t(x)\}. \quad (35)$$

On the scalar domain, all terms are piecewise affine. Their maximum is attained among the continuation knots, state nodes and cell boundaries; both adjacent actors are included at a boundary. The exact rational maximum supplies a valid all-state

allowance without an additional verification mesh. Both conventional interpolation and the nearest-node neural ablation receive this improved account. Thus the experiment does not contrast the new bound only against a deliberately loose baseline. The old separated bound remains recorded, and the smaller of two bounds for the same returned policy may be used.

8.4. *Matched construction, policy and certificate frontiers*

The new design was fixed in the repository before execution. It retains the non-linear continuous-innovation economy (27), horizons 2,4, prices 1,4, and all six resolutions 16,32,64,128,256,512. Three methods and three isolated timing repetitions produce 36 services and 216 rungs. Each rung restarts from primitives. The two cone methods have identical critics and node-action arrays; their deployed selectors differ. The spline constructs its own labels. All methods receive the same exact integration and outward numerical arithmetic.

The declared targets are $1/4$, $1/8$ and $1/16$. Every unsuccessful rung and every capped failure is retained. Internal prefix clocks include target construction, witness or nearest-cell compilation, certificates and durable checkpoints; process clocks additionally include imports and the common non-economic warm-up. Runs use one selected CPU and one numerical-library thread, with method order rotated across repetitions and economic cells. Frequency is not controlled. These are local measurements, not a hardware-independent work theorem. Primitive counts and rational storage are recorded separately from clocks, and are not called complete floating-point or bit-operation counts.

All three methods certify 12/12 services at $1/4$ and at $1/8$, and 6/12 at $1/16$. The latter failures are all horizon-four services and are retained. The witness bound is strictly tighter than the nearest-node neural ablation in all 24 distinct economy/resolution comparisons, and tighter than the strengthened spline in 23 of 24. The exception is $T = 4$, $p = 1$, $N = 16$, where the witness and spline bounds are approximately 2.06039 and 2.03515. The declared dyadic targets have the same first-crossing resolution for all three methods in every economic cell. Thus the certificate improvement does not yield an earlier first crossing on this particular ladder. The

TABLE VII

FINAL ALL-STATE POLICY-LOSS BOUNDS AT $N = 512$

T	p	Witness neural	Nearest neural	Spline
2	1	0.034308	0.050092	0.039641
2	4	0.035017	0.051637	0.040957
4	1	0.064387	0.093593	0.066369
4	4	0.065720	0.097039	0.069979

Bounds are rounded upward in the records; displayed decimals are summaries, not the target test. All three repetitions have identical checkpoints. Both nearest-node comparators use the strengthened one-sided certificate.

spline is faster in all 30 common successful target/repetition clock comparisons; the witness construction is faster than the nearest-node neural ablation in all 30. These local clocks and tighter regret bounds establish neither policy-cost superiority nor a general neural speed advantage. The comparisons isolate the effect of carrying the action witness and of using a joint policy certificate. They do not reclassify earlier capped services or claim that the new selector was previously deployed. In particular, an improvement shared by the strengthened nearest-node ablation cannot be attributed solely to witness selection. A tighter regret certificate does not identify the sign of a direct policy-cost contrast. The existing direct expected-cost intervals and adverse precision findings remain unchanged.

9. STATE-DEPENDENT CONSTRAINTS AND FEASIBLE WITNESSES

Investment opportunities often depend on current capital, collateral or remaining capacity. The common-action construction above does not by itself authorize transporting a node's action to a state where that action is inadmissible. We now incorporate a feasible repair into the same neural Bellman construction. The original controlled economy, rather than a relaxed control problem, remains the comparison object.

Use the invariant state domain, numerical-query conditions and common evaluation functional of Section 7. Replace its common action set by nonempty compact sets $A_t(x)$ satisfying

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad (36)$$

where d_H is Hausdorff distance in the action norm. The primitive inequalities (17) are now required for pairs of feasible state–action points; neither costs nor dynamics need to be evaluated at infeasible actions. Assume effective finite k_t -nets of $A_t(x_{t,i})$ and a feasible Borel repair $\Gamma_{t,i}(x) \in A_t(x)$ of each stored action $a_{t,i}$, with

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (37)$$

The nonnegative ζ_t is a certified displacement allowance, not permission to violate feasibility. Exact nearest-point selection gives $\zeta_t = 0$ when available. The effective repair and feasible-net procedures are additional computational inputs, whose work must be charged.

Starting with $L_T = L_g$, define

$$A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}, \quad (38)$$

$$D_t^Q = C_{a,t} + \beta_t F_{a,t} L_{t+1}, \quad (39)$$

$$L_t = A_t^Q + D_t^Q K_t^A.$$

At each state node, minimize the certified queries over its own feasible action net. Retain the original minimizing action with its numerical label $y_{t,i}$. Form the same cone network $f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}$. Let $i_t(x)$ attain this minimum and return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$.

THEOREM 10—Feasible-witness transport: *Under these conditions and the Bellman comparison hypotheses of Section 2, the constructed continuation is L_t -Lipschitz, and the returned policy is feasible and measurable. Uniformly on the original state domain,*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t^Q k_t + e_t + 2L_t h_t, \quad (40)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t^Q \zeta_t.$$

Consequently its loss relative to the optimum over all admissible adapted policies is at most

$$\sum_{t < T} B_t \{D_t^Q k_t + 2e_t + 2L_t h_t + D_t^Q \zeta_t\} + B_T(2e_T + 2L_g h_T). \quad (41)$$

No optimal continuation is supplied to the label generator. The bound concerns the policy after feasible repair, not the possibly infeasible array of unmodified node actions.

PROOF: The primitive inequalities imply a modulus A_t^Q in the state and D_t^Q in the action for feasible pairs. Given an action at x , the Hausdorff condition supplies an action at z within $K_t^A \|x - z\|_1$. Comparing the two objectives and reversing the roles of x, z shows that $\mathcal{T}_t f_{t+1}$ is L_t -Lipschitz. The feasible node net therefore gives $-e_t \leq y_{t,i} - \mathcal{T}_t f_{t+1}(x_{t,i}) \leq D_t^Q k_t + e_t$. The cone argument proves the first enclosure. For the attaining index, transport between the two feasible pairs and (37) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t^Q \zeta_t.$$

The cone equals $f_t(x)$. Apply Proposition 7 and the unchanged terminal enclosure. Q.E.D.

This result closes a constructive route for a class of endogenous constraints within the original model. It does not assume that all feasible correspondences have a finite Hausdorff modulus. With $s = \varepsilon / \sum_{t=0}^T B_t$, sufficient nonterminal allocations are $e_t \leq s/8$, $D_t^Q k_t \leq s/4$, $L_t h_t \leq s/8$ and $D_t^Q \zeta_t \leq s/4$. Together with the earlier terminal allocation, they imply finite termination at tolerance ε when all listed operations are effective. Zero coefficients impose no restriction. When exact repair is available, the larger query allowance of Corollary 9 is again sufficient. Setting $K_t^A = \zeta_t = 0$ recovers the common-action witness theorem.

9.0.0.1. A capacity-constrained investment rule. For two nonnegative state coordinates on $[0, 1]^2$, let $A(x) = [0, (x_1 + x_2)/2]$. This is a state-dependent capital-capacity constraint with $K^A = 1/2$ in the ℓ^1 state norm. The repair $\Gamma_i(x) = \min\{a_i, (x_1 + x_2)/2\}$ is feasible, satisfies (37) with $\zeta = 0$, and has an elementary affine-ReLU representation for each fixed witness. Thus changing capacity changes the actor and the constructive state modulus through an explicit term $D^Q/2$. The index selector remains a finite comparison procedure. The supplement provides exact rational checks for this example; these are checks of the theorem and admissibility contract, not an additional multidimensional performance study.

Neither feasible-net construction nor repair is free. With $J_{t,i}$ node actions, target-query work is summed over $\sum_i J_{t,i}$ rather than a common $K_t J_t$. Each deployed query also incurs the selected repair's work and storage. Under imperfect state acquisition,

repair must be certified feasible at the true state, or throughout the acquisition set for a robust implementation. An error allowance for the objective cannot substitute for that feasibility condition.

10. ECONOMIC APPLICATIONS AND CONCLUSION

The original applications concern policy-dependent recursive utility, endogenous preference specifications, temporal selves, controlled diffusion problems, and dynamic games. Their complete economic statements and proofs are retained without alteration. The centered and transport bounds explain how an NBO can support economic counterfactuals when changes in future primitives are controlled. The policy theorem explains what must be verified for the actual policy. Neither statement supplies an equilibrium or transversality condition absent from the application.

The nonlinear investment evidence adds an executable link from primitives to learned continuations and directly certified policies. Its retained-policy diagnostic demonstrates that a failed tight certificate can reflect the resolution of verification rather than the quality of the deployed policy. Its direct coupled comparison answers a different question—the relative cost of two policies—without confusing separate regret bounds with a method contrast.

Neural Bellman Operators thus organize construction, certification, economic reuse, and resource accounting around the policy actually returned. Their scientific assessment requires all of these objects. Flexible conventional methods remain important competitors; numerical activation is not a work saving, and a fixed catalogue is not a general reliability theorem. Theorem 5 and Corollary 6 now supply a separate deterministic reliability statement for an explicitly constructed neural backend. They do not retroactively prove convergence of the earlier optimizers. Preserving those distinctions makes the original economic program testable while retaining its full theory and applications.

APPENDIX A: READING AND PRESERVATION MAP

The present main article is the authoritative R46 exposition, extending the complete R45 manuscript. The active technical supplement contains complete proofs of its policy, retained-policy, paired expectation, and constructive-neural results.

The response maps every B1–B10 and M1–M10 comment to a mathematical result, an execution, or an explicitly unsettled comparative requirement. All active R45 labels, including the retained R44 labels, are preserved. The response distinguishes the latest supplemental review of the incomplete R43 snapshot from the earlier substantive R44 report. The new witness construction, execution allowance and matched frontier are stated and proved in this revision.

The current source tree also materializes the complete previous R41 article, supplement, and economic-applications companion without changing their bytes. They preserve the full controlled-diffusion, recursive-preference, endogenous-preference, temporal-self, game, and historical numerical developments. Their presence is preservation of those results under their original hypotheses, not new empirical validation across all applications. The current argument and new constructive proofs can be read without selecting another revision branch. Original R42 capped-service outcomes, R44 postselected diagnostics, and the predeclared R45 constructive catalogue are separately identified. No earlier review or revision is overwritten.

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